



CLOUD COMPUTING ARCHITECTURE :-

* A cloud computing architecture is a combination of components required for a cloud computing service. In other words, it refers to the components and sub-components required for cloud computing.

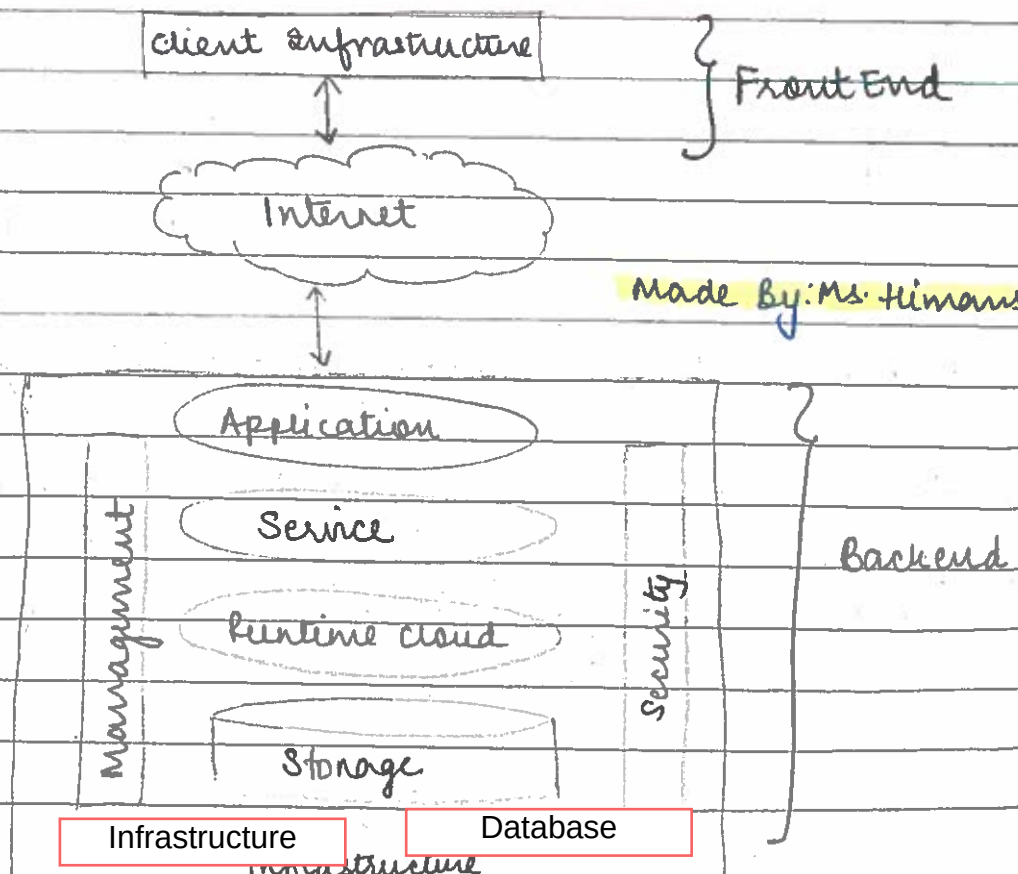
* The cloud computing architecture is broadly classified into 2 components: Front end and Backend.

↓
consists of client
part of CC system

↓
consists of implementation
part of CC system.

Made By :- Ms. Himanshi

* The subcomponents of CC architecture includes many different subcomponents like client infrastructure, applications, services, runtime clouds, storage spaces, management & security.





The cloud architecture is divided into 2 parts:-

- 1) Front End
- 2) Back End.

1) Front End:-

- * It contains of user interface used by the client to interact with the cloud.
- * It contains all the client side interfaces and applications that are required to access cloud platform.
- * It includes web browsers like chrome, Firefox, Opera etc, clients and mobile devices.

Made By:- Ms. Himanshi

2) Backend:-

- * It is used by service providers.
- * Here all the resources are managed that are required to provide cloud computing services.
- * Here by resources we mean, huge amount of data, storage, security mechanisms, virtual m/c, servers etc.

The important components of Architecture as explained below:-

① Client Infrastructure:- It is a front end component that provides a GUI. It helps users to interact with the cloud.

Made By:- Ms. Himanshi

② Application:- In backend, application refers to a s/w to which client accesses or any platform that a client wants to access.

③ Service:- It manages which type of service you can access according to the client's requirement. Three cloud computing services are:- SaaS, PaaS, IaaS.

- ④ Runtime cloud:- It offers the execution and runtime environment to the virtual m/c.
- ⑤ Storage:- It is one of the most important component as it provides a huge amount of storage capacity in the cloud to store and manage data.
- ⑥ Infrastructure:- It includes H/w & s/w components like servers, n/w devices, virtualization s/w & other resources needed for cloud computing model.
- Made By:- Ms. Himanshi
- ⑦ Management:- This component manages other components like application, service, cloud storage, infrastructure etc. It also establishes coordination between them.
- ⑧ Security:- Security in the backend refers to implementing different security mechanisms for secure cloud systems, resources, files and infrastructure to the end user.
- Made By:- Ms. Himanshi
- ⑨ Internet:- It acts as a bridge or medium between frontend & backend. It allows you to establish the interaction and communication b/w frontend & backend.

Benefits of CC Architecture:-

- * making overall cloud computing system simpler.
- * improves data processing requirements
- * helps in providing security
- * provides scalability.
- * encourages remote working & promotes team collaboration.

CLOUD COMPUTING SERVICE MODEL :-

CLOUD COMPUTING SERVICE DELIVERY MODEL :-

cloud computing can be defined as the practice of using a n/w of remote servers hosted on the internet to store, manage and process data, rather than a local server or PC.

companies offering such kinds of CC services are called cloud providers and typically charge for CC services based on usage.

made by: Ms. Himanshi

Grids and clusters are the foundation for CC.

* CC helps in rendering several services according to roles, companies etc. Cloud computing services are delivered on 3 basic models mentioned below :-

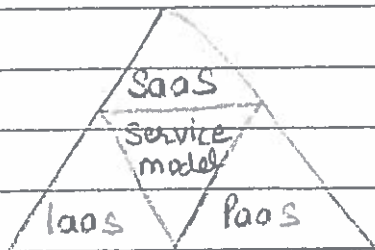
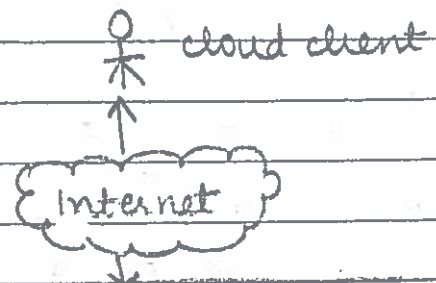


Diagram-1



SaaS: Business Apps Facebook, Youtube, Email
PaaS: Development tools, DB's, framework Microsoft Azure, Google App Engine
IaaS: Virtual m/c, servers, storage AWS

Diagram-2



① SaaS :-

- * It stands for Software as a Service.
- * Type of cloud service delivery used by the customers.
- * It is a way of delivering services and applications over the Internet.
- * Instead of installing and maintaining s/w, we simply access it via the internet, freeing ourselves from the complex h/w & s/w mgmt.
- * Maintenance of H/w & s/w is done by the vendor so, it removes the cost of maintenance.

Eg:- Gmail, Facebook, Games, Dropbox, Cisco-Webex, Office-365

Made By: Ms. Himanshi

- * In SaaS, customers neither need to install anything on their premises nor have to pay considerable upfront costs to purchase the s/w and the required licenses. Customers simply access the application website, enter their credentials and billing details and can instantly use the application. Made By: Ms. Himanshi
- * The SaaS model is appealing for applications serving a wide range of users and that can be adapted to specific needs with little customization.
- * SaaS applications are usually Multi-tenant i.e. it is a one to many s/w delivery model where one application is shared across multiple users.
- * Multitenancy is a feature of SaaS as compared to traditional s/w's allows providers to centralize and sustain the effort of managing large H/w infrastructure, maintaining and upgrading applications transparently to the users and optimizing resources by sharing the costs among the large user base.

Some core characteristics of SaaS are :-

- 1) Available over Internet
- 2) Maintained by the vendors or service providers.
- 3) Cost effective (pay as per use)



- 4) Available on demand.
- 5) Can be scaled up or down anytime according to our need.
- 6) Works on shared model i.e. one s/w is used by multiple clients.
- 7) s/w are automatically upgraded or centrally managed.
- 8) It is an integrated solution delivered on the contract, which means provided as promised.

Advantages of SaaS :-

Made by: Ms. Himanshi

- ① Cost Effective :- Pay only for what you use.
- ② Reduced time :- Users can run directly from web browsers without needing to download & install any s/w. This reduces the time spent in installation and configuration.
- ③ Accessibility :- We can access from anywhere.
- ④ Automatic updates :- Customers can rely on service providers to automatically perform updates.
- ⑤ Scalability :- Acquire or release resources as per requirement.

Made by: Ms. Himanshi

Disadvantages of SaaS :-

1) Dependence on Internet Connectivity :-

Since all services are internet based, so a stable internet connection is must to function properly.

2) Security concerns :-

They are responsible for maintaining the security of the data stored on their servers but still very risky.

3) Limited control over data :-

SaaS providers may have access to users data which is of great concern.

4) Limited Customization :- Not as customizable as on-premises s/w, meaning that users may have to work within the constraint of provide

② PaaS:-



DATE _____
PAGE _____

- * It stands for Platform as a service.
- * It is used by developers
- * It provides a platform and environment (i.e. runtime) to allow developers to build applications and services over the internet.
- * It offers development and deployment tools required to develop an application independent of H/W.
- * PaaS services are hosted on the cloud and accessed by the users via web browsers. **Made By:- Ms. Himanshi**
- * A PaaS provider hosts the H/W & S/W on its own infrastructure and frees the user from having to install in house H/W & S/W to develop or run a new application.

Foreg:- To conduct an annual day function, you will have 2 options: create a venue or rent a venue but function is same.

Made By: Ms. Himanshi

- * In PaaS, the consumer does not manage or control the underlying cloud infrastructure including n/w, servers, operating systems or storage but has control over the deployed applications and possibly configuration settings for the application hosting environment.
- * PaaS constitute the middleware on top of which applications are built.
- * Developers design their system in terms of applications and are not concerned with H/W, operating systems and other low level services.
- * Developers generally have the full power of programming languages such as Java, .Net, Python or Ruby with some restrictions to provide better scalability and security.
- * In this case, the traditional development environments can be used to design and develop applications, which are then deployed on the cloud by using the API's exposed by the PaaS provider.

Eg: Force.com, Longjump, IBM smart cloud, Google App Engine, Windows Azure

Advantages of PaaS:-

- ① Simple and convenient for users:-
It provides much of infrastructure and other IT services, which users can access anywhere via a web browser.
- ② Easy deployment of web applications Made by: Ms. Himanshi
- ③ Cost effective:- pay as per use.
- ④ Scalability is allowed.
- ⑤ No s/w management.

Disadvantages of PaaS:-

- 1) Limited control over infrastructure:-
PaaS providers typically manage the underlying infrastructure and take care of maintenance and updates but this can also mean that users have less control over the environment and may not be able to make certain customizations.
- 2) Limited flexibility:- Made by: Ms. Himanshi
PaaS solutions may not be able to accommodate certain types of workloads or applications, which can limit the value of the solution for certain organisation.
- 3) Dependence on the provider:- users are dependent on the provider for the availability, scalability and reliability of the platform which can be a risk if the provider faces outages or other issues.

③ IaaS:

- * It stands for Infrastructure as a Service.
- * It is related to providing Infrastructure.
- * This service is used by system admin/network admin.
- * It provides us o/s, n/w, servers for developing the applications.
- * It provides access to fundamental resources such as physical m/c, virtual m/c, virtual storage etc.
- * IaaS is a form of cloud computing that delivers the fundamental compute, n/w & storage resources to the consumer on demand, over the internet and on a pay as you go basis.
- * It is also known as HaaS/Hardware as a Service.
- * Typically, IaaS is a service where infrastructure is provided as outsourcing to enterprises such as n/w equipment, devices, database and web servers.
- * IaaS has greater control than PaaS & SaaS.
- * IaaS customers pay on a per-user basis, typically by the hour, week or month. Some providers also charge customers based on the amount of virtual m/c space they use.

Made by:- Ms. Himanchi

Eg: AWS, Oracle clouds, Google clouds
VmWare

Advantages of IaaS:-

- ① Cost Effective:- eliminates capital expense and reduces ongoing cost & IaaS customers pay on a per user basis, typically by the hour, week or month. Made by:- Ms. Himanchi
- ② Website hosting:- Running websites using IaaS can be less expensive than traditional web hosting.
- ③ Security:- The IaaS Cloud Provider may provide better security than your existing s/w.
- ④ Maintenance:- There is no need to manage underlying resources as done by IaaS Provider



Disadvantages of IaaS:-

- ① Limited access : It may not be accessible in certain regions and countries due to legal policies
- ② users are responsible for securing their own data & applications
- ③ limited control

Thus the options within IaaS umbrella offerings range from single servers to entire infrastructures including fire devices, load balancers, database & web servers

Made By: Ms. Himanshi

The main technology to deliver IaaS is H/W Virtualization i.e. one or more m/c configured and interconnected on top of which applications are installed

Applications	}	Applications	}	Applications	}
Data		Data		Data	
Runtime		Runtime		Runtime	
Middleware		Middleware		Middleware	
O/S		O/S		O/S	
Virtualization	} others manage	Virtualization	}	Virtualization	}
Servers		Servers		Servers	
Storage		Storage		Storage	
N/W		N/W		N/W	

SaaS

PaaS

IaaS

TYPES OF CLOUD / DEPLOYMENT MODEL :-

A deployment model defines how and where the cloud services are hosted and who can access them.

* The cloud deployment model identifies the specific type of cloud environment based on ownership, scale and access along with the cloud's nature and purpose.

* The location of the servers you are utilizing and who controls them are defined by a cloud deployment model.

Made by: Ms. Hlmanshi

* It specifies how your cloud infrastructure will look, what you can change and whether you will be given services or will have to create everything.

* There are 4 different types of cloud :-

1. Public

2. Private

3. Hybrid

4. Community

1) Public cloud :-

* It is open to all to store and access information via internet.

* Pay as per use

Made by: Ms. Hlmanshi

* Systems and services are easily accessible to general public.

* This type of cloud is used for B2C (Business to consumer) interactions. So here the computing resource is owned, governed and operated by government, an academic or business org.

* The public cloud makes it possible for anybody to access systems and services.

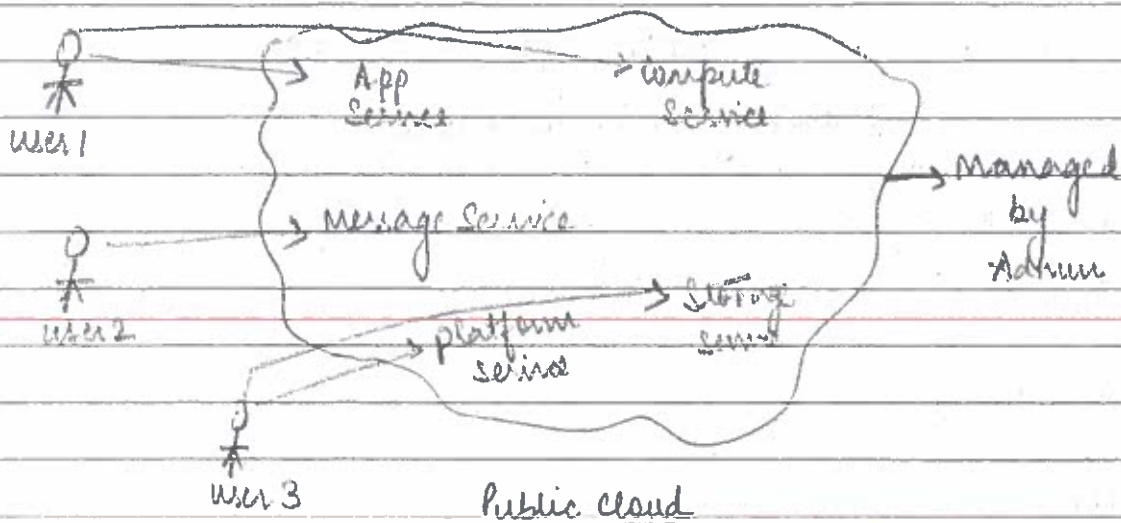
* The public cloud may be less secure as it is open to everyone.

* The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the customer.

* In this arrangement, storage backup and retrieval services

are given for free, as a subscription or on a per user basis.
 For eg: Amazon EC2, IBM, GCP

The fundamental characteristic is Multi-tenancy where several users share same resources.



made by: Ms. Himanshi

Advantages :-

- 1) Cost effective as it follows pay as per use.
- 2) High Scalability / Dynamic scalability to fulfill your company's need, on demand resources are accessible as per need. eg: Gmail
- 3) Location independent
- 4) The maintenance work is done by service provider not customer

Disadvantages :-

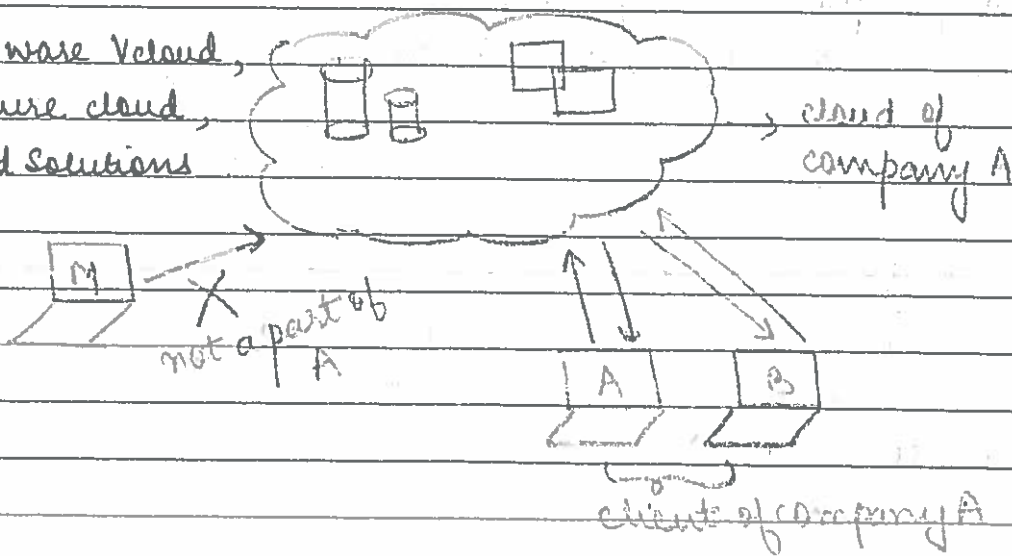
- 1) Less secure because resources are shared publicly.
- 2) Less customizable as compared to private cloud as it is accessed by many public so it can't be customized according to personal requirements.

So, they are the distributed systems containing one or more connected datacenters on top of which the specific services are implemented.

2.) Private cloud:-

- * Sometimes called as internal / corporate cloud.
- * Services accessible within an organisation i.e. belongs to specific organisation.
- * It can be managed by organization itself or 3rd party also
- * Private cloud is an exact opposite of public cloud.
- * The basic difference b/w public & private is ^{how} you handle H/W
- * Private cloud refers to the ability to access systems and services within a given border / organisation. Made by: Ms. Himanshi
- * The cloud platform is implemented in a cloud based secure environment that is protected by powerful firewalls and under the supervision of an organisation's IT department

Exeg: VMware Vcloud,
Cisco Secure cloud,
Dell cloud Solutions



Advantage of Private cloud:-

- ① High Security:- Here security concerns are less since data does not flow out of a private infrastructure.
- ② Privacy:- Only authorized people can access data.
- ③ more Customizable:- A private cloud allows a company to tailor its solution to meet its specific need.

Disadvantage of Private cloud:-

- ① Area of operation is limited / restricted area of operation.
- ② Scalability is limited.
- ③ Skilled people are required.

Made by: Mr. Himanshu

③ Hybrid Cloud:-

- * With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud cost savings. Made by: Mr. Himanshu
- * Hybrid clouds are the combination of features of public cloud (non-critical activities) or private cloud (critical Activities).
- * Public clouds have huge capability to serve the needs of multiple users and Private clouds are the perfect solution to keep the processing of information within the premises thus protecting from security threats. So a hybrid cloud could be an interesting opportunity for taking advantage of both clouds.

Hybrid clouds allow the use of existing IT infrastructure, keeping sensitive information within the organization, and can easily grow or shrink by adding or removing external resources when needed. Security concerns are then limited only to the public part of the cloud.



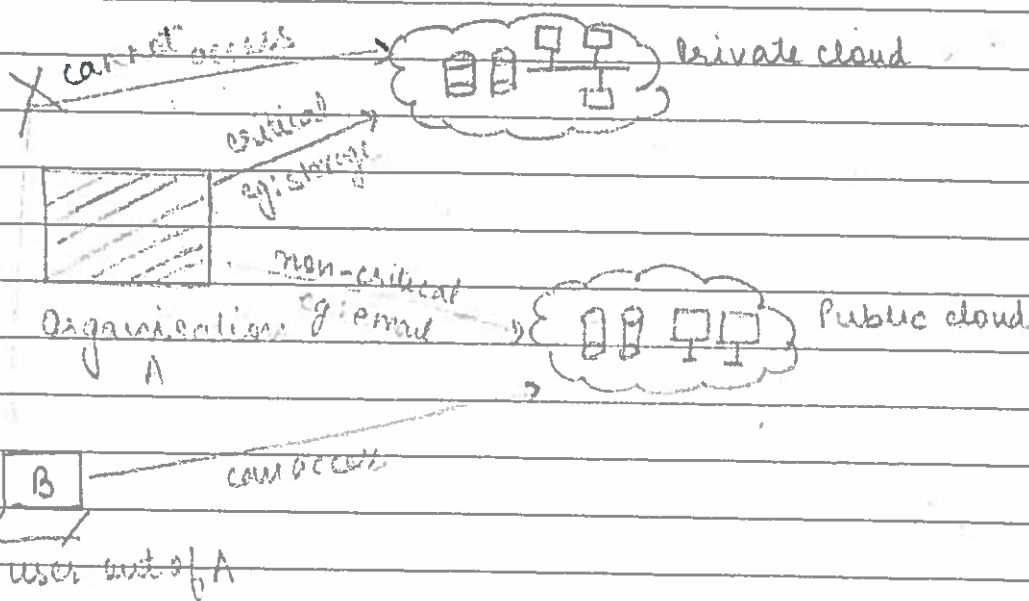
PAGE _____

* Hybrid clouds allow exploiting existing IT infrastructure, maintaining sensitive information within the premises and naturally growing & shrinking by provisioning external resources and releasing them when needed. Security concerns are then only limited to the public portion of the cloud.

* This technique in which resources are temporarily leased for the time required and then released. This practice is also called as cloud bursting.

For eg:- Azure stack, IBM cloud satellite, Oracle cloud.

Made by: Ms. Himanshi



Advantages :-

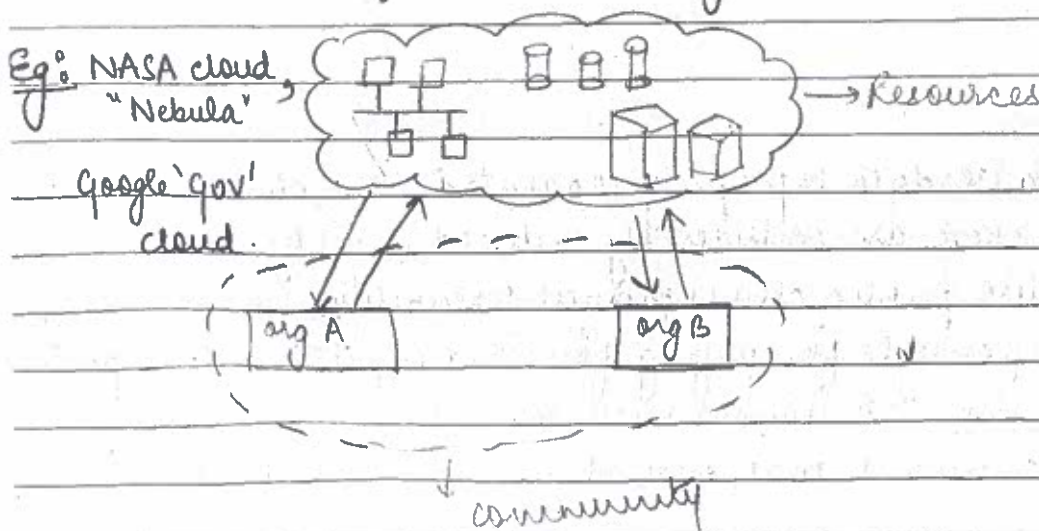
- ① Security: Because data is properly separated, the chances of data theft by attackers are considerably reduced. Made by: Ms. Himanshi
- ② Cost:- Because public clouds provide scalability, you will only be responsible for paying for extra capacity you require.
- ③ Flexibility & Control: Businesses with more flexibility can design personalized solutions that meet their particular needs.

Disadvantages of Hybrid :-

- 1) Hybrid clouds are difficult to manage as it is combination of both public & private cloud. so it is complex.
- 2) Network configuration issue due to the complexity of the cloud.
made by: Ms. Himanshi
- 3) Security on the part of public cloud.
- 4) Dependency on IT internal infrastructure.

F) Community cloud :-

- They fall between the category of public & private cloud.
- It allows systems and services to be accessible by a group of organisations for shared concerns like ^{Security} ~~security~~, compliance.
- It is created by integrating the services of different clouds to address the specific needs of a community, industry or business.
made by: Ms. Himanshi
- It is generally managed by a third party or by the combination of one or more org in the community.



* candidate sectors for community clouds are:- Media Industry, Healthcare Industry, Scientific Research Industry.

Advantages:-

- ① Cost effective: It is cheaper than private cloud as multiple companies share the bill which lowers the cost.
- ② sharing of resources among the companies
- ③ More secure than public but less secure than private

Made by:- Ms. Himanshi

Disadvantages:-

- ① We need to be careful in storing data [Data Segregation]
- ② Responsibility delegation among organisations

Economics Of The Cloud :- / CLOUD COMPUTING ECONOMICS :-

Made by:- Ms. Himanshi

* The main drivers of cloud computing are: Economy of scale and simplicity of software delivery and its operation.

* The biggest benefit of this phenomena is financial as this is pay as you go model offered by the cloud service providers. Also this model is highly scalable and simple

* In particular, cloud computing allows:-

- Reducing capital costs associated to the IT infrastructure.
- Eliminating the depreciation or lifetime costs associated with IT capital assets.
- Replacing the licensing with subscriptions.
- Cutting down the maintenance and administrative cost

* capital cost is the cost ~~incurred~~ occurred in purchasing an asset that is useful in the production of goods or rendering of services. capital costs are generally one-time expenses that contribute over a long term to generate profit.

At present time, it does not matter which type of business it as every enterprise will have an IT department. Hence

have an capital cost associated with the IT resources

Made By: - Ms. Jumanishi

* In the case of IT capital costs, the depreciation cost is represented by the loss of value of hardware overtime and the aging of s/w products which need to be replaced because new features are required.

Before the advent of cloud computing, the budget spend in IT infrastructure and s/w constituted a significant expense for medium and large enterprise. Many enterprises who own a datacenter, suffers several operational costs in terms of maintenance, electricity and cooling. Also, additional costs are occurred in maintaining an IT department and support centers.

Made By: Ms. Jumanishi

* After the cloud computing, these costs have significantly reduced or simply disappeared according to its advantages.

* Such a model has changed big one-time costs called capital costs into regular smaller payments called operational costs by renting the infrastructure and paying for software through subscriptions. Cloud computing also reduces management and maintenance costs. There is little or no need for administrative staff to take care of the cloud system because most of it is handled automatically by the cloud provider.

maintenance cost. There is no or limited need for having administrative staff taking care of cloud infrastructure.

* With the cloud computing, enterprise uses s/w applications on a subscription basis and there is no need of any licensing fee because the property of the s/w remains to the provider.

* The following are the three different strategies on which the pricing models for subscription are based.

a) Tiered Pricing :- In this, cloud services are offered in several tiers and each tier offers a fixed computing specification like CPU type, speed and memory that have a different cost per hour.

Made By: - Ms. Himanshi

b) Per-Unit Pricing :- In this, the source of revenue for cloud provider is determined in terms of units of specific services such as data transfer and memory allocation. This model is used, for eg: GIGABYTE where customers pay according to RAM/hr units for the servers deployed in GIGABYTE cloud.

c) Subscription based Pricing :-

This model is used mostly by users who are paying a periodic subscription fee for the usage of the s/w.

Made By: - Ms. Himanshi

~~So all of the above makes possible the conversion of
IT capital costs into operational costs as the cost of buying
is turned into interest by leasing or subscribing.~~

THE CLOUD REFERENCE MODEL:-

cloud computing supports any IT-service that can be consumed as a utility and it is delivered through the n/w i.e. Internet.. It is a blueprint or a map which tells us how cloud computing works.. The cloud computing reference model is an abstract model that characterizes and standardizes the functions of a cloud computing environment by partitioning it into abstraction layers and cross-layer function. This reference model groups the cloud computing functions and activities into five logical layers and three cross layer functions.

Made by:- Ms. Hemanshi

The five layers are:-

- (1) Physical layer (2) Virtual layer (3) Control layer
- (4) Service Orchestration (5) Service layer.

Each of these layers specifies various types of entities that may exist in cloud computing environment such as compute system, n/w devices, storage devices, virtualization s/w and so on. It also describes the relationships among these entities. ~~like floors in a building where every floor has its job.~~

Made by:- Ms. Hemanshi

The three cross-layer functions are: business continuity, security and service management. Business continuity and security functions specify various activities, tasks and processes that are required to offer reliable and secure cloud services to the customers. Service management function specifies various activities, tasks and processes that enable the administration of the cloud infrastructure and services to meet the provider's business requirements and consumers expectations.

~~these are like services that run through the whole building from bottom to top.~~



The layers are described below:-

① Service layer:-

* Consumers interact and consume cloud resources via this layer

* Entities that operate at this layer: Service Catalog and Self-Service portal

* Functions are:-

menu list of
all available services

↓
users can choose
and manage
own cloud

(i) Store information about cloud services in service catalog and presents them to the customers.

(ii) Enables customers to access and manage cloud services via a self-service portal.

made by: Ms. Himanshi

② Service Orchestration layer:-

* Entities that operate at this layer: Orchestration s/w.

* Functions are:-

(i) Provides workflows for executing automated tasks.

(ii) Interacts with various entities to invoke provisioning tasks.

* This layer works like an orchestra which uses special s/w to coordinate tasks.

③ Control layer:-

* Deployed either on virtual layer or on physical layer.

* Entities that operate at this layer: control s/w.

* Functions are:-

(i) Enables resource configuration.

made by: Ms. Himanshi

(ii) Enables resource pool configuration and resource provisioning.

(iii) Execute requests generated by service layer.

(iv) collaborates with the virtualization s/w and creating virtual resources and optimizing utilization of resources.

④ Virtual layer :-

- * Deployed on the physical layer.
- * Entities that operate at this layer: Virtualization software, resource pools, virtual resources.
- * Functions are :-

- Abstracts physical resources and makes them appear as virtual resources enabling multitenancy
- Executes the requests generated by control layer.

made by: Ms. Himanshi

⑤ Physical layer :-

- * It includes all physical things like computer, routers, switches, HD etc.
- * Foundation layer of the cloud infrastructure.
- * Entities that operate at this layer: Computer systems, network devices & storage devices.
- * Functions are :-

- Executes requests generated by the Virtualization & control layer.

Cross layer functions are described below:

① Business continuity :- * It makes sure cloud services keep running.

- Specifies adoption of proactive and reactive measures to mitigate the impact of downtime.
- Enables ensuring the availability of services in line with SLA.
- Supports all the layers to provide uninterrupted services.

② Security :- * Protects entire cloud

made by: Ms. Himanshi

- Specifies adoption of Administrative mechanisms i.e. security & personnel policies, standard procedures to direct safe execution of operations and technical mechanisms like firewall, antivirus.
- Supports all the layers to provide secure services.
- * Deploys security mechanisms to meet GRC requirements

Governance Risk Compliance

③ Service management :-

* Specific adoption of Activities related to service portfolio management and service operation management.

(i) Service portfolio management :-

- Define the service roadmap, service features and service levels.
- Assess and prioritize where investments across the service portfolio are most needed.
- Establish budgeting and pricing.
- Deal with consumers in supporting activities such as taking orders, processing bills and collecting payments.

Made By: Ms. Himanshi

(ii) Service operation management :-

- * Helps in setting up and managing infrastructure and resources.
- * Helps in solving problems.
- * Helps in managing the capacity and availability of services.
- * Helps in following rules and standards.
- * Helps in checking and watching cloud services and their parts.

CLOUD COMPUTING REFERENCE MODEL

SERVICE MANAGEMENT	Service layer Catalog Portal	BUSINESS CONTINUITY	SECURITY
	Service orchestration layer Orchestration software		
	Control layer:- control software		
Service Portfolio Management	Virtual layer Virtualization S/W Virtual Resources	Backup Replication	Security mechanism
Service operation Management	Physical layer Compute Network Storage	Fault tolerance mechanisms	GRC

INTERNET AS A PLATFORM :-

The concept of using the internet as a platform refers to utilizing the web for various services and applications beyond mere communication.

Initially the internet was a medium for information sharing but over time it has evolved into a comprehensive platform for software development, business operations and service delivery.

made by: Ms. Himanshi

Key Features

- Scalable: Can easily handle many users and large amounts of data.
- Accessible: Can be used from anywhere with internet access.
- Interoperable: Different systems and applications can work together smoothly.
- Service oriented: Services are designed in a modular way, making them reusable and flexible.

Main Components

- Web Services: Online services that can be used over the Internet such as booking flights or accessing weather data.
- API's: Tools that allow different software systems to communicate & share data.
- Service Models: SaaS, PaaS, IaaS. made By:- Ms. Himanshi

Benefits

- Cost Savings: Reduces the need to buy expensive H/W & S/W.
- Flexibility: Easy to adapt and scale services as needed.
- Collaboration: Makes it easier for people to work together online.
- Innovation: Encourages quick development and deployment of new applications.



Challenges :-

- Security :- Ensuring data & applications are protected from cyber threats.
- Privacy :- Safeguarding user information and complying with privacy laws.
- Reliability :- Maintaining constant availability and performance of services.
- Compatibility :- Making sure different systems and applications can work together without issues.

Important Examples of use :-

Made By: Ms. Himanshi

- Online shopping : websites like Amazon and eBay.
- Social Media : Platforms like Facebook, Twitter and Instagram.
- Streaming Services : Netflix for videos and ~~Spotify~~ Spotify for music.
- Online Learning : Educational platforms like Coursera & Khan Academy.
- Business tools : like Salesforce and SAP.

Summary :-

Made By: - Ms. Himanshi

The internet as a platform has transformed how we create and use services and applications. It offers many benefits like cost efficiency, flexibility and innovate quickly but also have many challenges in security, privacy & reliability. Understanding these basics helps us use the internet more effectively & safely.

CLOUD INFRASTRUCTURE & / CLOUD COMPONENTS :-

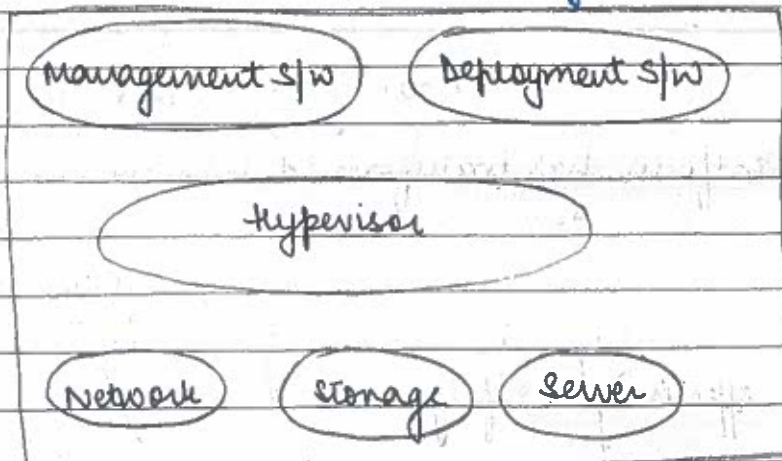
- cloud infrastructure which comes under the backend part of cloud architecture represents the H/W & S/W component such as server, storage, n/w.
- In backend, cloud infrastructure enables the complete cloud computing system.

Made by:- Ms. Himanshi

- * cloud infrastructure has more capabilities of providing the same services as the physical infrastructure to the customers. It is available for private cloud, public cloud and hybrid cloud with low cost, flexibility and scalability.

- * cloud infrastructure consists of servers, storage, n/w, cloud s/w, deployment s/w, platform virtualisation.

Made by:- Ms. Himanshi



components of cloud



* Hypervisor :- It is a firmware or low-level program that acts as Virtual Machine Manager. It allows to share the single physical instance of cloud resources between several tenants.

* Management s/w :- It helps to maintain and configure the infrastructure and monitors and optimizes resources, data, applications and services.

made by: Ms. Himanshi

* Deployment s/w :- It helps to deploy and integrate the applications on the cloud and helps in building a virtual computing environment.

* Network :- It is the key component of cloud infrastructure. It allows to connect cloud services over the Internet. It is also possible to deliver network as a utility over the Internet i.e. customizing n/w route & protocol.

made by Ms. Himanshi

* Server :- The server helps to compute the resource sharing and offers other services such as resource allocation and deallocation, monitoring the resources, providing security etc.

* Storage :- cloud keeps multiple replicas of storage. If one of the storage resources fails, then it can be extracted from another one, which makes cloud computing more reliable.

SOFTWARE PRODUCTIVITY IN THE CLOUD :-

The infrastructure required for developing & testing applications are different from the production environment like data security requirements are low.

- * Due to such variability in demand, where new development servers being required for each project, many become redundant after the application is released to production.

No Use

- * Setting up (provisioning and configuring) a new development environment can take a lot of time and effort in large companies because they have formal approval processes and IT management rules (like ITIL).

productivity in IaaS clouds

Made by: Ms. Himanshi

- * For Agile Business environment, where new projects are initiated regularly, delays can be impactful and become obstacle. So public cloud infrastructure in such cases can be fast, cost-effective as well as low risk.
- * During development & testing, designing application for good performance requires stress testing early which becomes difficult because of non-availability of such an environment for testing purpose.
- * Production capacity configurations are expensive and not so cost effective to invest in spare capacity for occasional performance needs. So using public cloud, production infrastructure can be provisioned on demand.



So we can ^{say} S/W development using public cloud infrastructure is better suited to support globally distributed development teams i.e. public clouds enable such teams to rely on central build infrastructure that is accessible over the Internet.

productivity in PaaS cloud

Made by:- Ms. Humaira

Let us reemphasize - that both Google & Microsoft PaaS platforms are unique in comparison to widely used development tools.

* The PaaS clouds provide development tools using which developers can create applications and both the above platforms built in scalability do not need to be concerned about how many servers are needed to run.

* Developers are supposed to write to the platform API's and scaling will be managed automatically by the cloud.

So if cloud platforms are able to factor out scalability and make it a feature of the platform than a lot of improvement in the productivity of large scale systems can be made.

ECONOMICS OF PRIVATE CLOUD:

Humaira

* The important driving factor for cloud economics is the way H/W capacity is planned for and purchased.

* From the perspective of an end user, cloud computing gives an illusion of a potentially infinite capacity with the ability to rapidly leverage additional capacity when needed and pay only for the one consumed.

Using the public cloud, obviates the need to plan ahead the peak loads and converts fixed cost to variable cost that change with actual usage thereby eliminating wastage. In contrast, provisioning for peak capacity is necessary within private datacenters and without adequate scaling there is limited opportunity to adjust capacity ~~across~~ across different applications so most datacenters operate at ridiculously low average utilization of 5-20%.

Private clouds come with an assumption of poor utilization for an in-house datacenter. *Kumarshi*

* It is discussed that using Virtualization and automatic provisioning may enable the inefficiencies of private cloud disappear and the benefits of cloud computing could be achieved on premise.

Following is the further cost analysis of the economics of private clouds in comparison with public clouds.

Let d be the computing demand of i^{th} application running at time t , so the aggregate demand will be given as

$$D(t) = \sum d_i(t)$$

The peak demand will be $D_{\max} = \max D(t)$. *Kumarshi*

The value of $D_{\max}$ and $D(t)$ depends upon the operational profiles of the applications.

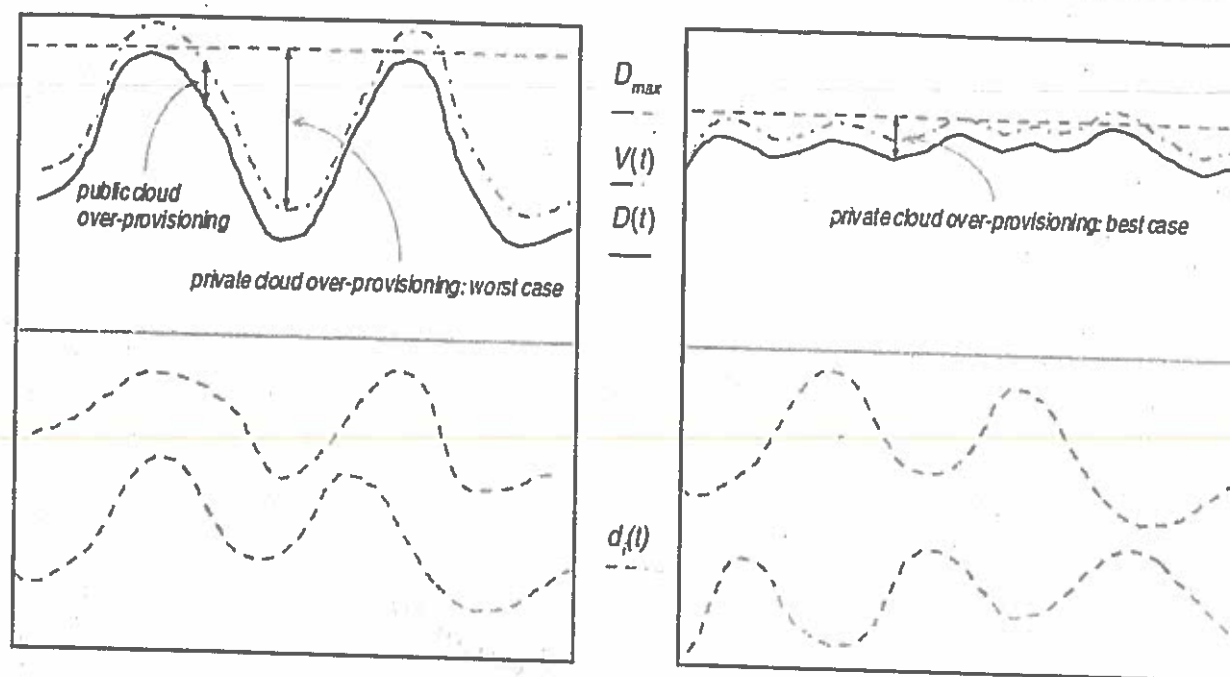


FIGURE 6.1. Over provisioning in private clouds

↓
if the peaks and troughs of different applications are highly correlated

$D(t)$ can vary from the peak

Kumarshi

↓
if the peak and trough of different applications are little or not correlated on the average and for large no of applications, $D(t)$ can be smooth and close to D_{max} .

Kumarshi

Now let S be the time delay to provision extra resources on demand. So if provisioning rate is far smaller than the rate of change of demand, then S one must keep provisioning the peak capacity D_{max} at all times.

So our analysis shows that even after improving utilizations at data center, because of critical applications which consume significant resources and need to be provisioned for peak loads, the benefits are rarely utilized. So over provisioning of resources will always be required so cost will always be more in case of private clouds.

ECONOMIES OF SCALE: PUBLIC VS PRIVATE :-

till now we have realized that public cloud services can be cheaper than private cloud.

* From the perspective of cloud providers, the advantages enjoyed are:-

1) Purchasing H/W, capacity, storage is 3 to 7 times cheaper if bought in the scale of hundreds and thousands of servers.

Amazon

2) Cloud providers can share the cost of server management among a large number of servers. They can also reduce these costs by using a lot of automation to handle tasks automatically.

3) Cloud providers take advantage of saving money by building their data centers in places where electricity is cheap and produced a lot, like Idaho and Washington in the US.

Amazon

4) Cloud providers make better use of their servers by serving many users at the same time using the same resources. This is done by creating an instance of it

number of users across the available capacity.

So it is very important to understand that the leading cloud providers Google and Amazon, developed these capabilities for other businesses so, it was very little marginal investment to be made for a new business model.



APPLICATION AREAS OF CLOUD COMPUTING :-

There are various applications of cloud computing as discussed below :-

(i) Business Application :-

Every organization requires the cloud business application to grow their business. For eg :- Salesforce : provides tools for commerce, sales etc; PayPal : for safe payments.

(ii) Data Storage & Backup Application :-

We can store files, data, images, audios & videos (eg: google drive)

(iii) Educational Applications :-

Online distance learning platforms are provided. Eg: Google Docs, classroom, Teams etc.

(iv) Entertainment Applications :-

Eg: Online Games, Video Conferencing App.

(v) Art Application :-

It offers art applications to quickly design attractive cards, booklets & images. Eg: Canva

(vi) Social Applications :-

It allows a large no. of users to connect with each other.
Eg:- Facebook, Instagram, Twitter etc.

COMPUTING PLATFORMS & TECHNOLOGIES

1) CLOUD COMPUTING PLATFORMS :-

cloud computing applications develop by leveraging platforms and frameworks. Various types of services are provided using a single platform or a combination of them. Some widely used cloud platforms are:-

① Amazon Web Services:- (AWS)

AWS offers cloud IaaS services ranging from virtual compute, storage and networking. AWS is mostly known for its compute and storage on demand services namely Elastic Compute Cloud (EC2) and Simple Storage Service (S3).

EC2 provides users with customisable virtual H/W that can be used as the base infrastructure for deploying computing systems on the cloud. EC2 instances are deployed either by using the AWS console or by using the web services API available for several programming languages. S3 is organised into buckets; these are containers of objects that are stored in binary form and can be enriched with attributes. Users can store objects of any size, from simple files to entire disk images and have them accessible from everywhere.

② Google AppEngine :-

Google AppEngine is a scalable runtime environment mostly devoted to executing web applications. These take advantage of the large computing infrastructure of Google to dynamically scale as the demand varies over time. AppEngine provides both a secure execution environment and a collection of services that simplify the development of scalable



and high-performance web applications.

③ Microsoft Azure :-

Microsoft Azure is a cloud operating system and a platform for developing applications in the cloud. It provides a scalable runtime environment for web applications and distributed applications in general. Applications in Azure are organised around the concept of 3 roles:

- Web role: designed to host a web application.
- Worker role: to perform workload processing.
- Virtual m/c role: provides a virtual env where the computing stack can be fully customized including the O.S.

④ Hadoop :-

Apache Hadoop is an open source framework that is suited for processing large data sets on commodity hardware. Hadoop is an implementation of MapReduce, an application model developed by Google, which provides two fundamental operations for data processing: map and reduce. The former transforms and synthesizes the i/p data provided by the user, while the latter aggregates the o/p obtained by the map operations. Hadoop provides the runtime env and developers need only to provide the input data and specify the map & reduce functions that need to be executed.

⑤ Force.com and Salesforce.com :-

Force.com is cloud computing platform for developing social enterprise applications. The platform is the basis of Salesforce.com — a SaaS solution for customer relationship mgmt. Force.com allows creating applications by composing ready-to-use blocks. The platform provides complete support for developing applications: from the design to defining rules and U.I.

(ii) CLOUD COMPUTING TECHNOLOGIES :-

We briefly review five core technologies that played an important role in the realization of cloud computing. These are:-

- 1) Distributed system
- 2) Virtualization
- 3) web 2.0
- 4) service oriented computing
- 5) utility computing

1) DISTRIBUTED SYSTEM :-

A distributed system is a collection of independent computers that appear to its users as a single coherent system.

The primary purpose of distributed systems is to share resources and to utilize them better. This concept is true in case of cloud computing where this concept is taken to the extreme & resources are sent to users.

2) Virtualization :-

Simulate

Virtualization is another core technology for cloud computing. It is essentially a technology that allows creation of different computing environments. These environments are named as virtual, because they simulate the interface that is expected by a guest. The most common example of virtualization is hardware virtualization. This technology allows simulating the H/W interface expected by an operating system. H/W virtualization allows the co-existence of different SW stacks on top of same H/W.



3) web 2.0 :-

The web is the primary interface through which cloud computing delivers its services. web 2.0 brings interactivity and flexibility into web pages, which provide enhanced user experience by gaining web-based access to all the functions that one normally found in desktop applications. These capabilities are obtained by integrating a collection of standards and technologies such as XML, AJAX, web services and others. web 2.0 applications are extremely dynamic as they improve continuously and new updates and features are integrated at a constant rate. Eg: Google maps, Google Docs, Youtube, Facebook, Twitter.

4) Service - Oriented computing :-

Service oriented

Service orientation is the core reference model for cloud computing system. This approach adopts the concept of services as main building blocks of application and system development. Service oriented computing supports the development of rapid, low-cost, flexible, interoperable applications and systems.

A service represents a self-describing and platform component that can perform any function ranging from simple to complex. A service is said to be loosely coupled, reusable, programming language independent. Loose coupling allows services to serve different scenarios more easily. Service oriented computing introduces 2 important concepts :-

- (i) Quality of service :- which identifies a set of functional & non functional attributes to evaluate the behaviour of service-like response time, reliability etc
- (ii) SaaS :- which is a new delivery model for applications. These deliver software services based solutions across WAN's on subscription & rental basis.

3) Utility-oriented computing:-

It defines a service provisioning model for compute services in which resources such as storage, compute power, applications and infrastructure are packaged and offered on a pay-per-use basis. This is based on the idea of providing computing as the utility like water, power, electricity.